



WEARABLE SYSTEM FOR MONITORING POSTURE AND HUMAN MOVEMENTS

MS4 19/05/2026



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Motion Suit

Our project involves developing a suit equipped with sensors that measure body posture and detect anomalies



PERFORMANCE



Ubuntu (Locust)



Mobile Hotspot
(Network Routing)



Mac (API + DB in Docker)

BASELINE



1 USER

~1.5 to 2.0 RPS

Driven by execution loop
speed

BASELINE



250 USERS

223.62 RPS

0.00% Failure Rate
Optimal Postgres pooling
and ASGI async handling

SATURATION



1000 USERS

24 to 32 RPS

22.29% Failure Rate
Thrashing state due to
network queue saturation

LATENCY TESTS

ENDPOINT		MEAN	95TH PERCENTILE
	GET /health	126ms	440ms
	GET /api/v1/users/me	426ms	1500ms
	GET /api/v1/me/gamification	441ms	1500ms

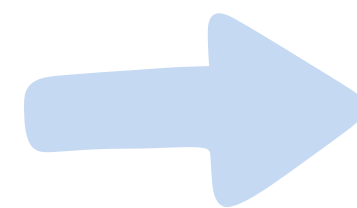
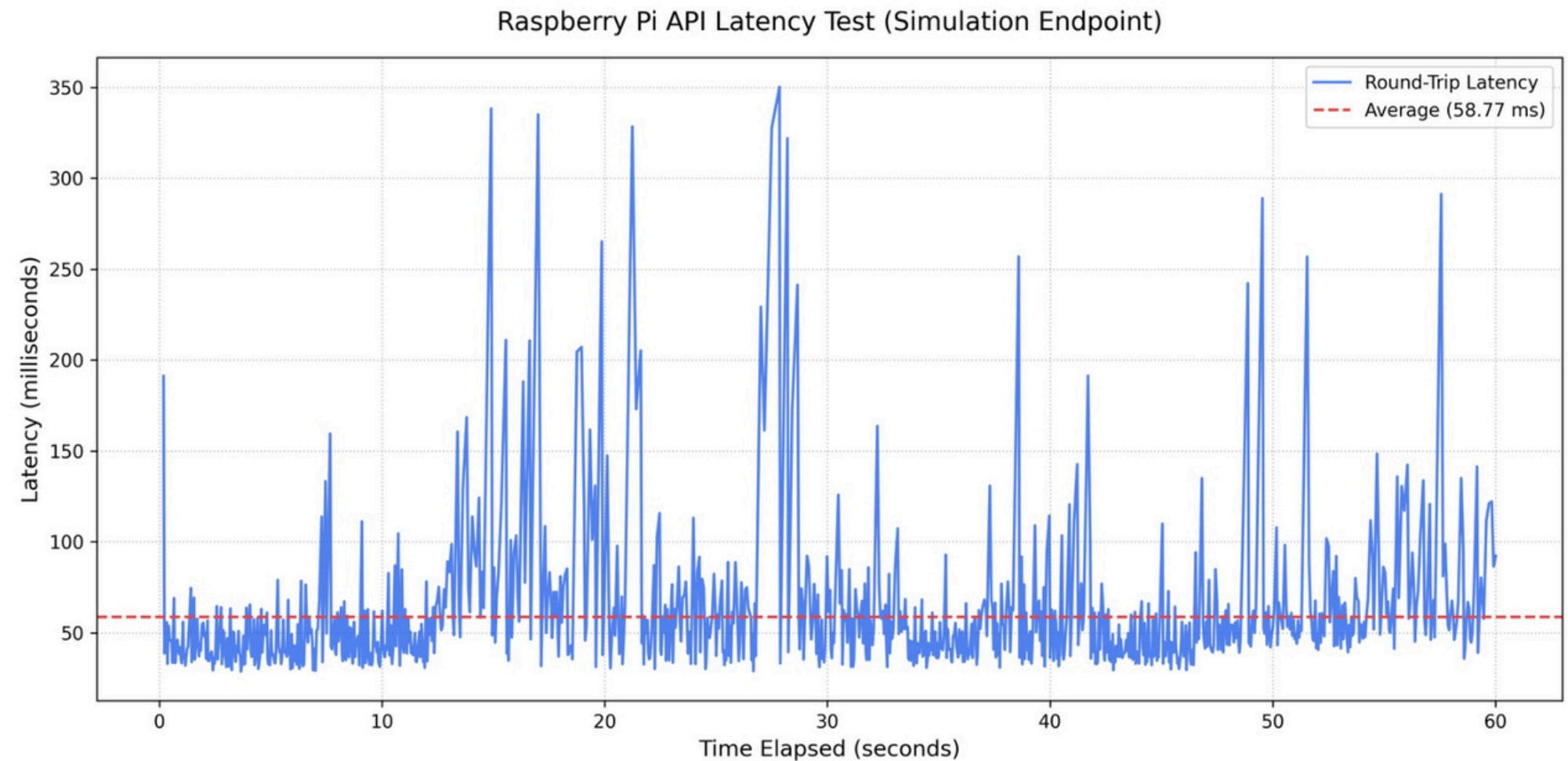
LATENCY TESTS

Single-User Dedicated Benchmark:

Comparative latency isolation without concurrent network load (1-to-1 suit architecture)

Execution Latency Breakdown:

Raspberry Pi Gateway: Average latency scales due to ARM architecture constraints.



58.77ms

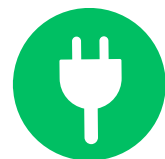
N+1 PROBLEM

Write Path Atomicity

Solved preference race conditions using a native PostgreSQL UPSERT (ON CONFLICT DO UPDATE)

Proposed Data Scaling

- Deploying compound time-series indexes on TimescaleDB for active sensor streams
- Implementing an In-Memory Cache middleware for static alert rules to achieve sub-5ms read latency



IMPLEMENTED



IMPLEMENTED



DESIGNED



ROADMAP

Database Connection Tuning

Expanded SQLAlchemy pool (pool_size=50, max_overflow=100) to eliminate connection starvation

N+1 Query Mitigation Plan

Identified lazy loading risks in the gamification router; resolved via a blueprint using joinedload() $O(N) \rightarrow O(1)$

SCALABILITY

Boundary Condition Stress Test:

Forced system saturation at

1000 concurrent users

System Degradation:

12.06s main response time

32.98 throughput RPS

Stabilized Failure Rate at 22.29%:

1174 failed requests

Dominant Error Category: RemoteDisconnected (Remote end closed connection without response).

SCALABILITY

Software Layer Integrity

The Python core and DB connection pools remained fully operational. Marginal Internal Server Errors (500) occurred solely due to transactional database deadlocks under extreme write contention.

Physical Virtualization Bottleneck

86.9%

macOS kernel_task consumption to manage network I/O interrupts inside the Docker Desktop VM.

This triggered severe CPU Starvation for the API container.

SCALABILITY

Vertical Scaling Boundary

4 workers led to resource starvation → failures rose to 12.55%

Horizontal Scaling Proof

Native stateless microservices containerized for independent multi-node deployment

Edge-to-Cloud Roadmap

Edge Layer

Raspberry Pi processes real-time posture alerts locally (100% offline)

Cloud Layer

Managed databases (AWS RDS / Timestream) to offload physical host storage

Async Ingestion

Event-driven Message Broker (Kafka / RabbitMQ) to buffer heavy sensor stream

SECURITY

Identity, sessions and permissions

Identity and Session Management

Access Control

Attribute-Based Access Control

SECURITY

Identity, sessions and permissions

Identity and Session Management

- Delegated Identity (Keycloak)
- Secured all protocols
 - Not only REST but also the websocket (for the 3D model)

Access Control

Attribute-Based Access Control

SECURITY

Identity, sessions and permissions

Identity and Session Management

Access Control

- **Vertical: Principle of least privilege**
 - FastAPI dependencies
 - Fail-fast architecture
- **Horizontal**
 - Users only have access to their own data

Attribute-Based Access Control

SECURITY

Identity, sessions and permissions

Identity and Session Management

Access Control

Attribute-Based Access Control

- Researchers only have access to participant's data if the participant has consented

SECURITY - OWASP top 10

1

Broken Access control

- Vertical access control (role-based)
- Horizontal access control (IDOR Prevention)
- Protocol-agnostic Security (Websocket is also secured)

2

Cryptographic Failures

- Delegated Hashing to Keycloak (Industry standard hashing algorithms)
- Token cryptography (verify RSA signatures)

3

Injection

- SQLAlchemy as our ORM
- Horizontal access control (IDOR Prevention)
- Protocol-agnostic Security (Websocket is also secured)

SECURITY - OWASP top 10

4

Insecure Design

- ABAC

5

Security Misconfiguration

- Containerization
- Environment variables
- Gitleaks

6

Vulnerable and Outdated Components (Supply Chain)

- Performed dependency auditing (both frontend and backend)
- CI/CD automation for dependency evaluation

7

Identification and Authentication Failures

- Delegation to Keycloak
- Stateless sessions (short-lived, stateless JWTs)

SECURITY - OWASP top 10

8

Software and Data Integrity Failures

- CI/CD with SAST
- Pydantic schemas for shape, type and integrity validation of JSON payloads

9

Security Logging and Monitoring Failures

- API logs for security-related events (when a user tries to snoop data from another user)

10

Server-Side Request Forgery (SSRF)

- Secure internal data fetching from the Raspberry Pi
- This URL comes from the server configuration (through .env)

ALGORITHM VALIDATION

Invariants

TEST SCENARIO:

Negative metrics (Weight/Height < 0) or age outside the range [18-120]

RESULT:

Automatic error 422

Competition errors

TEST SCENARIO:

Simultaneous start of two sessions for the same suit

RESULT:

Blocking via Race-Condition Check (400 Bad Request)

Transactional inconsistency

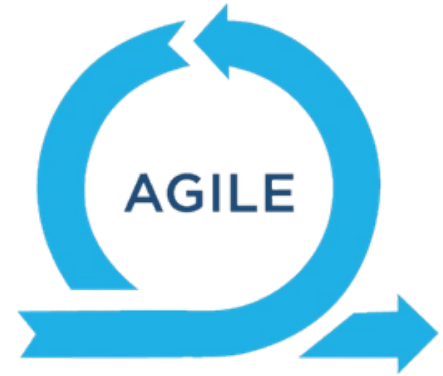
TEST SCENARIO:

Keycloak failed during onboarding

RESULT:

DB Rollback executed successfully

PROJECT MANAGEMENT



Agile Planning



Milestone-Driven
Sprints

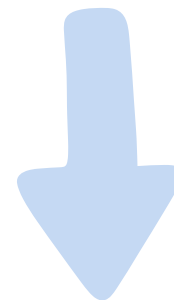


Hardware-dependent
tasks delayed until
sensors arrived

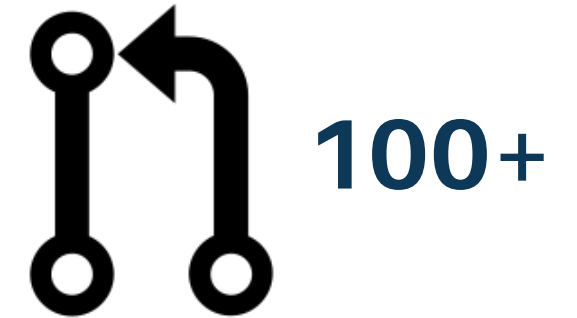


CI/CD

2 pipelines failures on main
branch



SSH errors during Azure
deployment



PR Review

WebSocket 404 Error
on production



Incorrect env variable
configuration

PROJECT MANAGEMENT

Workflow:



Pull request (example):

Feat/setup cd implementation #67

Merged **BeatrizF26** merged 5 commits into `merge/cd-and-vm-config` from `feat/setup-cd-test` 3 days ago

BeatrizF26 and others added 5 commits 2 weeks ago

Merge pull request [#54](#) from [PEI-MotionSuit/dev](#) Verified [2fa0986](#)

- [Add workflow template for testing](#) [f7d8aaa](#)
- [Update pipeline trigger to match new branch name](#) [1a0f793](#)
- [Fix workflow indentation format](#) [a970521](#)
- [added cache](#) [14db3bb](#)

All checks have passed
2 successful checks

- Backend CI Quality Gate / Test & SonarQube Scan ...** [Details](#)
- SonarCloud Code Analysis** Successful in 41s — Qu... [Details](#)

CALENDAR

May 19

MS4: System component validation

T-14

Done

Unit and Integration Testing

T-15

Done

Performance and Escalibility Testing

T-7

Done

Calibrate the suit

May 26

MS5: Demo + Poster for Students@DETI & Video

T-13

In Progress

Integration with a smartwatch

T-16

In Progress

Cable and Sensor Installation

T-17

In Progress

Poster and Video



<https://pei-motionsuit.github.io/microsite/>

THANK YOU!



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